

PharmaFlow AI

Business Document

Prescription Cost Optimization Agent for Insurance Companies

One-line pitch: *PharmaFlow AI helps insurance companies find expensive brand-name prescription claims that may have lower-cost generic options, estimate savings, and create doctor/patient review documents before any switch is made.*

1. The Problem: Insurance Companies Pay for Expensive Drugs Without a Clear Action Path

In the U.S. healthcare system, the person taking the medicine is usually not the person paying most of the bill. A doctor writes the prescription, the patient fills it at a pharmacy, and the insurance company or health plan often pays a large share of the cost. A Pharmacy Benefit Manager (PBM) helps manage drug coverage, pharmacy networks, formularies, prior authorization, and rebates.

This creates a business problem for insurance companies: they may pay a high amount for a brand-name medicine even when a cheaper FDA-approved generic version exists. The insurer can see the final paid claims in its data, but the data is usually spread across claims tables, NDC codes, drug names, quantities, doctor details, member IDs, pricing files, and formulary rules. Turning that raw data into a clear action plan is slow.

What users do today: Today, a pharmacy analyst would usually export claims into Excel or SQL, manually group brand drugs by NDC/name, cross-check possible generics against pricing files or formulary data, estimate savings, and then prepare separate notes for clinical or formulary review. This process is slow, inconsistent, and hard to repeat across thousands of claims every month. PharmaFlow-AI is useful because it turns this manual spreadsheet workflow into a repeatable claim-to-action pipeline.

Why PharmaFlow-AI: PharmaFlow-AI does not automatically tell a patient to switch medicine. Instead, it finds possible savings opportunities and packages them for human review. The insurer can then ask a doctor, pharmacist, or clinical team to review whether the patient can safely move from the brand drug to the generic alternative.

Doctor prescribes	Patient fills	Insurance pays	Agent finds generic option	Doctor/pharmacist reviews
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2. The User: The Paying Customer Is the Insurance Company

The main customer is not the patient. Patients benefit because they may pay lower copays and get clearer explanations, but patients usually do not control drug coverage rules. The insurance company is the user and buyer because it pays a large part of the prescription cost and has the power to review claims, change policies, work with PBMs, and coordinate doctor review.

Customer	Concrete User	Problem Today	Why They Pay
Primary: insurance company / health plan	Pharmacy analytics analyst, medical economics analyst, or pharmacy benefit director	Claims are available, but it is hard to quickly know which brand-drug claims have a cheaper generic option and which require clinical review.	If the plan saves even a small amount per member, it can reduce drug spend, improve margins, and potentially support more competitive insurance pricing.
Secondary: PBM or benefits team	Formulary manager or client service lead	They need a simple way to explain savings opportunities and review exceptions.	The tool creates a repeatable review process instead of one-off manual spreadsheet work.
Indirect: patients and doctors	Patient, physician, pharmacist	They may not see total plan cost or generic options at the right time.	They benefit from clearer, safer cost conversations, but they are not the main buyer.

Example business value: if an insurer finds that 500 members are using a high-cost brand drug and a doctor-reviewed generic option saves even \$40 per member per month, that is \$20,000 in monthly savings on one opportunity. Across many drugs and members, the savings can become meaningful enough to lower costs, improve the plan's financial position, and help the insurer serve more members.

3. Product Flow: What the Insurance Company Uploads and What the Agent Returns

The input is a pharmacy claims CSV, not prescription images. Insurance companies already analyze structured claims data. A typical row says which member filled which drug, the NDC code, the quantity, days supply, amount paid by the plan, amount paid by the member, pharmacy, and prescriber. PharmaFlow AI turns that spreadsheet into a decision-ready savings report.

Flow Step	Explanation
Step 1: Upload claims file	The payer analyst uploads a claims CSV with drug name, NDC, quantity, days supply, plan paid amount, member paid amount, member ID, prescriber, and fill date.
Step 2: Find brand-to-generic matches	The Librarian Agent maps brand drugs to possible FDA-approved generic equivalents using Orange Book and NDC/drug-name lookup logic.
Step 3: Estimate cost difference	The Auditor Agent uses NADAC unit cost as a public price benchmark and calculates estimated gross savings for each possible switch.
Step 4: Add clinical and access review	The Clinician Agent and Social Navigator Agent flag cases where switching may be risky, unclear, or hard for the member because of access/adherence constraints.
Step 5: Produce action package	The dashboard shows top opportunities and, on Gold, generates a four-document switch package for doctor, patient, pharmacist, and internal payer review.

4. Pricing and Packaging

The product is sold as monthly software to insurance companies and health plans. Pricing is based on how much workflow the insurer wants: high-level detection only, risk-aware review, or full member-level action packaging.

Plan	Price / Month	Best For	Included	Not Included
Bronze	\$500	Small plan doing a quick savings scan	Brand-vs-generic opportunity list; gross savings estimate; dashboard summary	No member-level view; no downloads; no clinical risk or access risk flags
Silver	\$1,200	Payer team that wants safer prioritization	Everything in Bronze plus clinical-risk and access-risk summaries, Recommend/Review/Do Not Switch bands, portfolio charts	No member-level details; no CSV export; no switch-package document download
Gold ★	\$2,500	Insurance company ready to act on findings	Everything in Silver plus member-level view, CSV export, and four switch-package documents: doctor review note, patient explanation, pharmacist outreach letter, and internal payer/formulary memo	No real-time rebate modeling unless added as enterprise integration

Why Gold matters: the insurer does not need to search its own database to find which member is using the brand drug. Member-level detail gives a single place to see the claim, candidate generic, estimated savings, clinical/access warnings, and the documents needed to start a human-reviewed switch workflow.

Gold Plan: The Switch Package

As part of the Gold Plan, PharmaFlow AI automatically generates a complete Switch Package with four ready-to-use documents, so the insurance company does not just see a savings opportunity, it also gets the paperwork needed to act on it.

1. Internal Utilization Management Memo

This is for the insurance company's internal pharmacy benefits and clinical review teams. It summarizes the member, current brand drug, proposed generic, expected savings, clinical/access risk, and the final recommendation.

2. Prescriber Clinical Review Letter

This is sent to the doctor or prescribing clinician. It explains the proposed generic substitution, shows the equivalence evidence, risk scores, and asks the clinician to approve or deny the switch.

3. Member Benefit Letter

This is sent to the patient/member. It explains in simple language that a lower-cost generic alternative may be used, that the prescriber has been notified, and that the member can request an exception if needed.

4. Pharmacy Network Alignment Notice

This is sent to the dispensing pharmacy. It tells the pharmacy which generic drug/NDC may replace the brand drug, confirms that prescriber approval is needed, and helps the pharmacy prepare for the member's next refill.

The four documents are important because cost savings only matter if the company can act. The package gives the insurer a doctor-facing note, a patient-friendly explanation, a pharmacist outreach letter, and an internal payer memo. This turns the agent from a dashboard into an operational workflow.

5. Economics: Does One Customer-Month Make Money?

Back-of-envelope assumption: one small insurance customer analyzes 10,000 pharmacy claims in a month. About 1,000 claims need deeper brand-to-generic review, and the payer downloads 150 switch packages. Most calculations are database lookups and deterministic Python formulas; the LLM is used mainly for fallback summaries and plain-English explanations.

Item	Bronze	Silver	Gold
Monthly price	\$500	\$1,200	\$2,500
Expected cloud + database cost	\$60	\$90	\$150
Expected LLM/token cost	\$1–\$5	\$3–\$10	\$10–\$30
PDF/export generation cost	\$0	\$0	\$25–\$75
Support + monitoring allowance	\$100	\$200	\$350
Estimated total cost to serve	\$161–\$165	\$293–\$300	\$535–\$605
Approx. gross margin	~67%	~75%	~75%

Token economics: if 1,000 analyzed opportunities use about 4,000 input tokens and 1,000 output tokens each, that is roughly 4M input tokens and 1M output tokens. Even at a low-cost flash model assumption of about \$0.10 per 1M input tokens and \$0.40 per 1M output tokens, the LLM bill is under \$1 for core analysis. The estimate above includes a higher buffer for retries, summaries, uploaded CSVs, and generated explanations. The important point is that PharmaFlow is not token-heavy because the expensive work is done by DuckDB lookups, pricing formulas, and structured outputs. Gold uses more output tokens because the four-document switch package creates longer doctor, patient, pharmacy, and internal payer explanations, but this is still bounded because documents are generated only for selected opportunities, not every claim.

Break-even logic: Gold only needs one or two meaningful switch opportunities to justify its price. For example, if the product helps the insurer safely review 100 members where the net saving is \$75/member/month, that is \$7,500/month in potential savings versus a \$2500/month Gold subscription. Larger plans can produce much higher savings because the same software cost is spread across more members.

Where the economics break: The economics break if the customer is too small, uploads too few claims, or uses the product only for reports without acting on any switch opportunities. For example, if a Gold customer pays \$2,500/month but only identifies \$1,000/month of realistic savings, the product is not justified. The model works best when the payer has enough monthly claim volume and enough brand-drug spend for even a small number of safe switches to exceed the subscription price.

6. Why the Technical Choices Fit the Business

Technical Choice	Business Reason
FastAPI + single web app	Keeps deployment simple for a class demo and for a small payer. One service can host the upload flow, dashboard, API, and document downloads.
DuckDB warehouse over NADAC + Orange Book	Insurance users need fast, repeatable answers. DuckDB gives cheap SQL lookups for pricing and equivalence instead of calling an LLM for every claim.
Four-agent pipeline	The work naturally separates into four jobs: find the generic, price the switch, check clinical risk, and check access/adherence. This makes the output easier to trust and explain.
Structured Pydantic outputs	Savings and risk fields are validated before they reach the dashboard. This reduces hallucination risk and makes the result more audit-friendly for healthcare users.
RAG	RAG helps answer unknown-drug questions, but it is not the core pricing engine. That keeps costs low and avoids using the LLM for sensitive calculations.
Gold switch-package artifacts	The business value is not just finding savings; it is helping the insurer act. Downloadable documents make the product useful inside real payer workflows.

These technical choices protect the business model because the agent does not use the LLM as the main calculation engine. DuckDB, Orange Book lookup logic, NADAC pricing formulas, and structured validation handle the high-volume work cheaply. The LLM is reserved for summaries, explanations, and document generation, which keeps token costs low and makes the \$500–\$2,500/month pricing profitable. Because the customer is a healthcare payer, the system must be explainable and reviewable. Structured outputs, deterministic pricing logic, and doctor/pharmacist review language make the agent safer than a black-box recommendation tool.

Note: PharmaFlow AI is decision support, not a medical prescriber. It says “requires doctor/pharmacist review,” not “switch this patient.” The class demo uses de-identified Aetna Claims data. A real customer version would need secure data handling, HIPAA-aware workflows, access controls, audit logs, and contract-specific rebate/formulary integrations.

Public source notes for demo data: FDA Orange Book for approved drugs and therapeutic-equivalence information; CMS NADAC for public drug acquisition-cost benchmarks. Model-cost numbers above are simple planning assumptions and should be replaced with the deployed provider's current contract pricing.